

4-Week Daily Internship Workflow

Development of an AI-Based Model to Detect and Predict Underwater Environmental Changes Using Water Quality Parameters

MODE	DURATION	INTERN DISCUSSION	MENTOR REVIEW
Online / Individual Execution	4 Weeks / 28 Days	Every 2 Days	Every 3 Days

Mandatory Individual Execution

Every intern must complete the entire project independently from scratch. Work must not be divided among team members. Intern discussions are only for comparing findings, identifying blockers and preparing mentor questions; code, models, experiments and final submissions remain individual.

Dataset & Hardware Scope

All required datasets will be provided before the end of Week 1. The mandatory project is software-only and no physical hardware integration or on-device deployment is required.

Week 1 | Environmental Analytics & Dataset Readiness

Meeting Protocol

Intern-only group discussion: every 2 days (30-45 minutes). Mentor review: every 3 days (45-60 minutes). On overlap days, the intern discussion happens first. Before each mentor review, every intern prepares a maximum 3-slide progress pack: (1) work completed, (2) results/findings, and (3) blockers + next 3-day plan.

Week 1 | Problem Understanding, Research & Dataset Readiness

Week 2 | Temporal Cleaning, EDA & Feature Engineering

Day	Focus	Individual Tasks	Daily Output	Meeting
1	Problem Understanding & System Definition	- Read the complete problem statement and rewrite it in your own technical words. - Identify the inputs (timestamped temperature, pH, dissolved oxygen, pressure and available related water-quality variables) and the required output/target (anomaly flags/scores and future environmental parameter values). - Draw the end-to-end software pipeline and define success criteria. - Prepare at least 8 technical questions or assumptions requiring clarification.	Problem Understanding Note; pipeline diagram; questions list	—
2	Literature Review - Domain + ML	- Review the core domain concepts for AI / Machine Learning / IoT Data Analytics / Underwater Environmental Monitoring. - Study 6-10 closely related methods, datasets or prior systems.	Literature review; comparison table; candidate-method list	Intern Discussion 1

Day	Focus	Individual Tasks	Daily Output	Meeting
		• Prepare a paper/method comparison table with data, approach, metrics, results and limitations. • Shortlist candidate methods including statistical anomaly baseline; Isolation Forest or comparable detector; regression/Random Forest/XGBoost forecasting; sequence model only where justified.		
3	Research Framing & Mentor Review	• Refine the research question and measurable outputs. • Prepare a maximum 3-slide individual progress pack: completed work, findings, blockers and next plan. • Confirm target definition, dataset structure, metrics and scope with the mentor. • Record mentor decisions and update assumptions.	3-slide progress pack; revised research framing; mentor notes	Mentor Review 1
4	Dataset Design & Quality Expectations	• Define expected files/columns/labels/targets and important metadata before using the final dataset. • List likely missing-data, annotation, imbalance, temporal or leakage risks. • Define what a correct train/validation/test split should preserve. • Prepare a dataset-risk checklist.	Dataset schema/structure note; risk checklist	Intern Discussion 2
5	Pipeline & Experiment Planning	• Design the reproducible workflow from raw data to final evaluation. • Define baseline method, advanced methods and the project-specific experiment. • Set tentative evaluation criteria: anomaly detection rate or precision/recall/F1 where labels exist; MAE/RMSE for forecasting; processing time. • Create experiment-log and repository/folder structure.	Pipeline flowchart; experiment plan; log template; folder structure	—
6	Pre-Dataset Readiness + Mentor Review	• Complete literature, methodology and data-readiness package. • Present proposed preprocessing, modelling and evaluation plan. • Confirm split strategy, metrics and constraints with the mentor. • Update Methodology V1 after feedback.	Methodology V1; updated experiment plan; mentor corrections	Intern Discussion 3 + Mentor Review 2

Day	Focus	Individual Tasks	Daily Output	Meeting
7	Dataset Receipt & Initial Inspection	• Preserve an untouched raw copy of the supplied dataset. • Inspect files/rows/images/videos/sequences, labels/targets and metadata. • Check missing/invalid data, duplicates, class/target coverage and obvious inconsistencies. • Generate first-look statistics or visualisations without irreversible cleaning.	Raw-data archive; initial inspection notebook; preliminary dataset report	—

Week 2 | Data Cleaning, EDA & Feature Engineering

Week 3 | Anomaly Detection & Forecasting Experiments

Day	Focus	Individual Tasks	Daily Output	Meeting
8	Temporal Data Quality	• Audit timestamps, missing intervals, duplicates, invalid sensor ranges and irregular sampling. • Create a sensor-wise data-quality table. • Identify drift/gaps and leakage risks. • Propose cleaning/resampling rules.	Sensor data-quality table; cleaning/resampling plan	Intern Discussion 4
9	Temporal Cleaning Review	• Present timestamp/data-quality findings. • Confirm missing-data and resampling strategy. • Define chronological train/validation/test splits. • Freeze approved cleaning plan.	3-slide progress pack; approved temporal split	Mentor Review 3
10	Environmental EDA	• Plot sensor trends, distributions and correlations. • Inspect temporal patterns and candidate anomalies. • Analyse cross-sensor interactions. • Record anomaly/forecast hypotheses.	Environmental EDA notebook; observations	Intern Discussion 5
11	Preprocessing Pipeline	• Implement cleaning, chronological splits and scaling. • Create reusable timestamp/sensor preprocessing. • Preserve raw and processed versions. • Document assumptions.	Processed Dataset V1; temporal pipeline	—
12	EDA /	• Present trends, candidate anomalies, splits and	Dataset V1; approved	Intern Discussion

Day	Focus	Individual Tasks	Daily Output	Meeting
	Preprocessing Review	leakage checks. • Defend cleaning/resampling choices. • Resolve mentor corrections. • Freeze Dataset V1.	pipeline	6 + Mentor Review 4
13	Temporal & Sensor Features	• Create lag, rolling, time-based and cross-sensor features where justified. • Ensure features are available at prediction time. • Compare raw vs engineered features. • Document feature definitions.	Feature-engineering notebook; feature comparison	—
14	Week-2 Peer Consolidation	• Compare EDA/anomaly hypotheses without code sharing. • Discuss temporal patterns and sensor quality. • Record one peer insight and one hypothesis. • Prepare Week-2 package.	Week-2 package; peer insight; hypotheses	Intern Discussion 7

Week 3 | Model Development & Comparative Experiments

Week 4 | Integrated Monitoring Analysis & Final Output

Day	Focus	Individual Tasks	Daily Output	Meeting
15	Anomaly Baseline + Mentor Review	• Implement a simple statistical/rule anomaly baseline. • Implement Isolation Forest or comparable ML anomaly detector. • Visualise detected anomalies. • Present and confirm evaluation strategy.	Anomaly baseline notebook; anomaly plots	Mentor Review 5
16	Forecasting Baselines	• Build persistence/simple regression forecast baseline. • Train Random Forest/XGBoost or comparable model. • Evaluate MAE/RMSE on chronological holdout. • Plot predicted vs actual values.	Forecast notebook; prediction plots; metrics	Intern Discussion 8
17	Advanced Anomaly / Forecast Model	• Train a stronger anomaly or sequence model only if justified. • Keep complexity proportionate to the dataset. • Track validation behaviour.	Advanced-model notebook; candidate results	—

Day	Focus	Individual Tasks	Daily Output	Meeting
		Save candidate model.		
18	Model / Method Comparison + Mentor Review	- Prepare one comparison table covering all methods built so far. - Compare performance using anomaly detection rate or precision/recall/F1 where labels exist; MAE/RMSE for forecasting; processing time. - Discuss overfitting, data limitations, complexity and failure patterns. - Shortlist the strongest candidate(s) for deeper experiments.	3-slide progress pack; method comparison; shortlisted candidates	Intern Discussion 9 + Mentor Review 6
19	Controlled Hyperparameter / Parameter Experiments	- Tune only shortlisted candidate(s) with a documented search strategy. - Change meaningful settings systematically and log every experiment. - Use validation data only for selection and keep the final test set untouched. - Save the best validated configuration.	Experiment log; tuned candidate(s)	—
20	Project-Specific Robustness / Ablation Study	- Conduct the planned special analysis: sensor/feature ablation, chronological validation and dashboard-style monitoring visualisation. - Evaluate the same metrics across relevant conditions or component variants. - Quantify the effect of the tested change/condition. - Interpret the result in terms of the research question.	Robustness/ablation table or notebook; research insight	Intern Discussion 10
21	Best Candidate Review	- Present baseline versus tuned results and the project-specific study. - Show representative successes and failures. - Discuss remaining weaknesses and unusual errors. - Get mentor approval for the Week-4 final direction.	Best-candidate checkpoint; approved Week-4 direction	Mentor Review 7

Week 4 | Edge Optimisation, Validation & Final Research Output

End-of-week gate: Project ready for individual final evaluation.

Day	Focus	Individual Tasks	Daily Output	Meeting
-----	-------	------------------	--------------	---------

Day	Focus	Individual Tasks	Daily Output	Meeting
22	Final Model / Pipeline Freeze	• Select the final method using validation evidence, robustness and appropriate simplicity. • Freeze preprocessing, split and evaluation methodology. • Save final model/pipeline artefacts and outputs. • Document the technical selection rationale.	Frozen final model/pipeline; reproducible artefacts	Intern Discussion 11
23	Software Performance / Final Validation	• Measure relevant runtime, inference/processing time and model size where applicable. • Run final validation on the held-out data using the frozen pipeline. • Compare final method against the strongest simple baseline. • Record accuracy/quality-versus-complexity trade-offs.	Final-validation notebook; performance comparison	—
24	Final Technical Review	• Present the final method, validation results and remaining questions. • Show the reproducible input-to-output workflow. • Confirm final error-analysis and reporting requirements. • Record mentor corrections.	3-slide progress pack; final review; correction list	Intern Discussion 12 + Mentor Review 8
25	Detailed Error / Failure Analysis	• Inspect worst predictions/outputs individually. • Organise recurring errors by relevant conditions, classes, ranges or scenes. • Use feature importance/visual explanations where appropriate. • Document limitations and likely failure conditions.	Error/failure-analysis notebook; limitation list	—
26	Final Software Package & Report Draft	• Clean repository and package preprocessing, model/pipeline and example inference/output workflow. • Write README and setup/run instructions. • Prepare final figures, tables and result summary. • Draft final report and presentation.	Final software package; README; report/PPT draft	Intern Discussion 13
27	Final Mentor Review	• Present near-final methodology, results, software artefacts and limitations. • Show repository, report draft and presentation draft. • Resolve final mentor corrections.	Near-final report/PPT; mentor correction list; technical sign-off	Mentor Review 9

Day	Focus	Individual Tasks	Daily Output	Meeting
		Freeze technical results after approved changes.		
28	Peer Mock Presentation & Final Submission	- Run the full individual presentation for the other interns without the mentor. - Test timing, clarity and ability to defend technical choices. - Verify repository links, dataset/model references and reproducibility. - Complete and submit the entire individual project package.	Final report; final PPT; repository; model/pipeline; outputs; future scope	Intern Discussion 14 / Mock Presentation

Week Final | Individual Submission Package

#	Mandatory Deliverable
1	Problem Understanding Note and Literature Review
2	Dataset Understanding / Quality Report
3	Preprocessing / EDA Notebook
4	Feature / Representation Preparation Notebook
5	Baseline Model / Method Notebook
6	Advanced Model / Method Experiments
7	Model / Method Comparison and Experiment Log
8	Project-Specific Robustness / Ablation / Error Analysis
9	Final Model / Software Pipeline and Output Artefacts
10	Clean Source-Code Repository with README
11	Final Individual Project Report
12	Final Individual Presentation and Future Scope